

SUDHIR POL

📍 Seattle, Washington 📞 +1 (930) 904-4785 ✉ sudhirpol522@gmail.com 🔗 [linkedin.com/in/sudhir-pol-a6b544179](https://www.linkedin.com/in/sudhir-pol-a6b544179) 🌐 [sudhirpol522.github.io](https://github.com/sudhirpol522)

Summary

Data Scientist with 3 years of industry experience at American Express, S&P Global, and Adobe building models that answer business questions and shipping them to production on AWS. Works across classical machine learning, NLP, and LLM systems, with a habit of defining the evaluation metric before the model and measuring impact in time saved and decisions improved.

Education

Indiana University

Master of Science in Data Science

Aug. 2024 - May 2026

Bloomington, United States

Rajarambapu Institute of Technology

Bachelor of Technology in Electronics and Tele-Communications

Aug. 2017 - May 2021

Sangli, India

Experience

Adobe

May 2025 - Aug. 2025

Machine Learning Engineer Intern

San Jose, California

- Built an LLM as Judge evaluation framework with LiteLLM to benchmark Gemini 2.5 Pro, GPT 4.1, and Claude 4.5 on multimodal chart quality tasks, raising offline F1 score from 79% to 92% by redesigning the scoring metrics with domain reviewers
- Automated a 48 hour manual evaluation workflow into a 30 minute pipeline using a LangGraph agent with Model Context Protocol over Adobe PDF Spaces, adding a human in the loop approval gate so stakeholders could review each plan before execution

S&P Global

Nov. 2022 - Aug. 2024

Data Scientist II

Hyderabad, India

- Designed and deployed a question answering API on AWS EC2 using a T5 transformer for question generation and answering over financial filings, with added logic to resolve current year figures from long documents
- Fine tuned and deployed LayoutLMv3 for token classification on AWS EC2, converting unstructured financial documents into structured fields that analyst teams could search and verify
- Deployed Llama 2 on SageMaker and AWS Lambda with chain of thought prompts to extract information from document pages, saving about 20 minutes of data collection time per analyst task
- Engineered a Java regex framework using named groups to extract varied structures from financial documents, replacing a paid third party vendor with an in house solution and saving about 10 minutes of analyst review time

American Express

Dec. 2021 - Nov. 2022

Data Scientist I

Gurgaon, India

- Designed the VIBE call quality model combining XGBoost with BERT based embeddings to score the sentiment of customer calls, containerized with Docker and used through the year as the basis for customer representative bonuses
- Built a risk classification model that sorted customer complaints into 3 risk levels using a naive Bayes classifier with TFIDF features, calibrated probabilities with isotonic regression, and deployed the model on AWS EC2
- Developed an LSTM demand prediction model mapping call transcripts to demand types, deployed with Flask on AWS EC2, so the business could answer high priority customer calls first
- Delivered a PySpark proof of concept using TFIDF and regex to detect missing offer disclosures in call transcripts, helping the business identify representatives giving customers incomplete offer information

Projects

MultiDocChat: Production RAG System | *FastAPI, LangChain, FAISS, Kubernetes, ArgoCD*

May 2025

- Built a production RAG system with FastAPI, LangChain LCEL, FAISS indexing, and MMR retrieval, using Gemini 2.5 Pro for semantic scoring and LangSmith for end to end tracing
- Implemented GitOps CI/CD with GitHub Actions, ArgoCD Image Updater, and Kubernetes manifests, achieving zero downtime rolling deployments with semantic versioning

NFL Big Data Bowl: Play Outcome Prediction | *Python, Scikit-learn, Pandas, NumPy*

Nov. 2020

- Framed yard gain prediction as a probabilistic problem, predicting the full cumulative distribution of outcomes rather than a point estimate, and calibrated the predictions with isotonic regression for reliable probabilities
- Benchmarked Logistic Regression, SVM, Random Forest, Gradient Boosted Trees, and XGBoost after extensive exploratory analysis, data cleaning, and football specific feature engineering on play level data

Writing

- The Math of AWQ: Protecting Salient Channels from the Inside Out *Jun. 2026*
- Demystifying GPTQ: From Lagrange Multipliers to Vectorized PyTorch *Jun. 2026*
- Speculative Decoding: Make LLM Inference Faster *Oct. 2025*

Skills

Languages: Python, SQL, PySpark, Java, C++

Machine Learning: Statistical Modeling, XGBoost, Random Forest, SVM, Deep Learning, NLP, Probability Calibration, Feature Engineering, Experiment Design, LLMs, RAG

Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, HuggingFace Transformers, LangChain, FastAPI, Flask

Cloud and MLOps: AWS (SageMaker, EC2, Lambda, S3), Docker, Kubernetes, MLflow, GitHub Actions, CI/CD